

Aprameya Satish

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Active Secret Clearance

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Applied machine learning and statistical signal processing for acoustics, RADAR/SONAR, and physiological sensing—with technical leadership across defense and health programs.

EDUCATION

- **Georgia Institute of Technology** Atlanta, GA
Ph.D. in Mechanical Engineering; GPA: 4.0 Aug 2015 – Dec 2020
- **Georgia Institute of Technology** Atlanta, GA
M.S. in Electrical and Computer Engineering; GPA: 4.0 Aug 2017 – Dec 2020
- **BMS College of Engineering** Bangalore, India
Bachelor of Mechanical Engineering Aug 2009 – June 2013

SKILLS

- **Languages:** Python, MATLAB, C/C++
- **Domain Knowledge:** Applied machine learning for acoustics, RADAR, SONAR, and health; statistical signal processing; spatial array processing; computer vision; acoustics and infrasound; elastic wave propagation; simulation and synthetic data generation

EXPERIENCE

- **Georgia Tech Research Institute** Smyrna, GA
Senior Research Engineer Dec 2020 – Present
 - **Cognitive Radar Modeling:** Technical lead for machine learning algorithm development on a new cognitive radar program at GTRI, sponsored by the Missile and Space Intelligence Center (MSIC).
 - **Physiological monitoring and health state estimation:** Key contributor to an ONR-sponsored, multi-million-dollar collaboration with Georgia Tech on military fitness monitoring and prediction (publications and a patent). Extending this line of work to health state estimation from longitudinal wearable data collected from athletes.
 - **Acoustic field testing and source characterization:** Led and supported acoustic field tests and post-test data analysis to characterize sources of interest, including signature extraction, annotation, and model validation against measured datastreams.
 - **SONAR imaging simulator:** Developing a SONAR imaging simulation capability to generate controlled synthetic returns for algorithm development, testing, and characterization of underwater sensing pipelines.
 - **Vibroacoustic Pipeline Leak Detection:** Technical lead for experiment design, data management, and statistical DSP/ML algorithm development for pipeline leak detection using vibroacoustic datastreams.
 - **Infrasound source localization and classification:** Built array-based localization pipelines, ML signature classifiers, and a synthetic infrasound signature modeling suite for targets of interest.
 - **SAR Automatic Target Recognition:** Key contributor to signal processing and deep-learning target detection for AFLCMC-sponsored SAR ATR.
 - **Computer-vision underwater target detection:** Technical lead for an internal program on ML-based underwater landmark detection from SONAR imagery for autonomous vehicle localization.
 - **UAV tracking using acoustic beamforming:** Led algorithm development for outdoor UAV tracking with spherical-array acoustic beamforming on a DARPA program.
 - **Infrasound and acoustic exposure analysis:** Collaboration with WRAIR (US Army), NSWC, and Raytheon on mitigating sound and infrasound exposure during weapons and blast training.
 - **Maritime Infrasound Monitoring System:** Coordinated wireless communications and signal processing for a realtime ONR infrasound beamforming dashboard.
 - **Internal Research Management:** Secured and managed approximately \$210k of internal (IRAD) funding for underwater navigation/threat detection with acoustics and machine learning for health.
- **Georgia Institute of Technology** Atlanta, GA
Graduate Research Assistant Jan 2018 – Jul 2018
 - **Auditory UAV based source localization:** Acoustic source localization in low-SNR environments using ground-mounted and MAV-mounted microphone arrays.
 - **Monochannel audio source separation:** Statistical ML framework for acoustic source/noise separation via latent variable decompositions with limited training data.
 - **Jet Noise analysis using NMF:** Non-negative matrix factorization analysis of jet-engine mode-switching combustion instability.
- **SpaceX** Hawthorne, CA
Propulsion Dynamics Intern May 2016 – Aug 2016

- **Merlin Engine Dynamics Analysis:** Analyzed Merlin Stage I and vacuum engine dynamics from pressure, accelerometer, and strain-gauge data; defined nominal operating envelopes via statistical analysis; automated rainflow fatigue evaluation on strain data.

Mercedes-Benz Research & Development India

Bangalore, India

Product Design Engineer

Jul 2013 – Jul 2015

- **Transmission Design:** Engineering design and analysis for Mitsubishi FUSO commercial vehicle transmissions, including component stress simulation, parametric CAD, and GD&T; led a transmission-case redesign for manufacturing and variant compatibility.
- **CAD Automation:** Initiated and led AutoGear, an automated CAD prototyping tool for rapid gear modeling and performance analysis, deployed across Daimler.

PATENTS

- Patent arising from the ONR-sponsored soldier health monitoring collaboration with Georgia Tech (application/issuance details to be confirmed).

PH.D. THESIS

A. Satish, *Analysis and Design of Passive Underwater Acoustic Identification Tags*. PhD thesis, Georgia Institute of Technology, 2020

REFEREED JOURNAL PAPERS

1. D. J. Lin, A. Satish, K. L. Richardson, S. An, C. O. Yaldiz, M. Buller, K. Driver, E. Atkinson, T. Mesite, C. King, O. T. Inan, and A. Medda, "Predicting soldier performance on structured military training marches with wearable accelerometer and physiological data," *IEEE Sensors Journal*, 2023
2. C. O. Yaldiz, M. Buller, K. L. Richardson, S. An, D. J. Lin, A. Satish, K. Driver, E. Atkinson, T. Mesite, C. King, M. Bursey, M. Galer, M. M. Stafford, M. N. Sawka, A. Medda, and O. T. Inan, "Early prediction of impending exertional heat stroke with wearable multimodal sensing and anomaly detection," *IEEE Journal of Biomedical and Health Informatics*, 2023
3. A. Satish and K. G. Sabra, "Passive underwater acoustic identification tags using multi-layered shells," *The Journal of the Acoustical Society of America*, vol. 149, no. 5, pp. 3387–3405, 2021
4. A. Satish, D. Trivett, and K. G. Sabra, "Omnidirectional passive acoustic identification tags for underwater navigation," *The Journal of the Acoustical Society of America*, vol. 147, no. 6, pp. EL517–EL522, 2020
5. A. Satish, B. Nichols, D. Trivett, and K. G. Sabra, "Passive underwater acoustic tags using layered media," *The Journal of the Acoustical Society of America*, vol. 145, no. 1, pp. EL84–EL89, 2019

SELECTED CONFERENCE PROCEEDINGS

1. S. Das, A. Satish, N. Breen, M. Fitzpatrick, and A. Medda, "Development of synthetic infrasound signatures for target classification using window of opportunity observations," *The Journal of the Acoustical Society of America*, 2025
2. W. N. Newcomb, S. Reddy, A. Satish, and A. Medda, "Time-frequency spectral optimization of underwater sonar acoustic returns for the purpose of characterization using spectral descriptors," *The Journal of the Acoustical Society of America*, vol. 156, no. 4-Supplement, pp. A25–A26, 2024
3. A. Spexet, N. Breen, A. Satish, and D. Alvord, "Intelligent summarization of aviation data for maintenance," in *2024 IEEE Aerospace Conference*, pp. 1–10, IEEE, 2024
4. A. Satish, E. Laubie, J. Casey, and T. Selee, "Automatic target chip extraction from sar images using cnns and cfar," in *Automatic Target Recognition Working Group Meeting*, National Geospatial Intelligence Agency, 2023
5. A. Satish and A. Medda, "Acoustic uav detection using spherical array beamforming," in *2022 Asilomar Conference on Signals, Systems, and Computers*, IEEE, 2022
6. A. Satish and A. Medda, "Time-frequency beamforming adaptation for micro aerial vehicle (mav) noise suppression and source localization," in *2018 52nd Asilomar Conference on Signals, Systems, and Computers*, pp. 884–888, IEEE, 2018

Additional ASA meeting abstracts on passive underwater acoustic markers and jet mode-switching (2017–2021).

PROFESSIONAL SERVICE

- Technical Co-Chair, IEEE SoutheastCon 2024 (*Aug 2022 – present*)
- Treasurer, IEEE Atlanta Chapter (*Feb 2022 – Feb 2025*)
- Reviewer, *Measurement* (*Dec 2020 – present*)
- Reviewer, Acoustical Society of America (*Dec 2020 – present*)
- Grant Committee Member, Melton Foundation (*Dec 2020 – present*)
- Senior Fellow, Melton Foundation (*Jun 2013 – present*)
- Grad Gala Operations and Logistics Chair, GT Graduate Student Association (*Mar 2020*)
- Treasurer, Acoustical Society of America GT Student Chapter (*Dec 2017*)
- President's Undergraduate Research Award (PURA) Grant Committee Technical Reviewer (*Jun 2016, Jun 2020*)
- Co-Organizer and Compliance Officer, TEDxBangalore (*Aug 2014*)

RECOGNITION

- Outstanding Service Award, IEEE Atlanta Section (*Oct 2024*)
- Outstanding Service Award, GT Graduate Student Association (*Mar 2020*)
- GTRI Best Internally Funded Project (IRAD) nominee (*Jun 2018*)
- Mercedes-Benz R&D Star Performer nominee (*Jun 2014*)